Georeferencing Report
76D BORRHÅLSKARTA.jpg

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Image file:      76D BORRHÅLSKARTA.jpg
Image size:      8544 × 5662 pixels
Sheet code:      76D
Sheet bounds:    X [100900..101700]  Y [76000..76500]

Detection grid:  8 cols × 4 rows  (27 GCPs detected)
Grid spacing:    100 m easting, 100 m northing
Pixel scale:     ~10.36 px/m (E)  ~10.39 px/m (N)

Source CRS:      EPSG:3152  (ST74, Stockholm 1938)
Target CRS:      EPSG:3006  (SWEREF99 TM)
Pipeline:        pyproj (PROJ)

Inter-consistency (affine fit):
  RMS residual:          0.484 m
  Max residual:          1.070 m
  Outlier threshold:     0.967 m (2×RMS)

Overall quality: GOOD
  (1 outlier(s): 20)
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Crosses detected by normalised cross-correlation template matching
Transformation equivalent to Lantmäteriet GTRANS (inv ST74 TM → UTM33 on GRS80)

GCP Locations — 76D BORRHÅLSKARTA.jpg

Inter-consistency — Affine Fit Residuals

GCP	Pixel	Line	ST74 X	ST74 Y	Res X (m)	Res Y (m)	Res (m)	Corr	Flag
1	2219.0	1115.0	101000	76400	-0.580	0.583	0.822	0.856	
2	3248.0	1115.0	101100	76400	0.090	-0.324	0.336	0.867	
3	4283.0	1114.0	101200	76400	0.181	-0.034	0.184	0.901	
4	5320.0	1115.0	101300	76400	0.078	-0.199	0.213	0.914	
5	6357.0	1118.0	101400	76400	-0.025	-0.171	0.173	0.776	
6	7393.0	1124.0	101500	76400	-0.031	0.146	0.150	0.894	
7	1183.0	2146.0	100900	76300	-0.701	0.231	0.738	0.871	
8	2213.0	2147.0	101000	76300	-0.128	0.067	0.145	0.844	
9	3242.0	2148.0	101100	76300	0.542	-0.096	0.550	0.714	
10	4278.0	2150.0	101200	76300	0.536	-0.164	0.560	0.846	
11	5317.0	2152.0	101300	76300	0.240	-0.233	0.334	0.887	
12	6355.0	2155.0	101400	76300	0.040	-0.206	0.210	0.869	
13	7391.0	2158.0	101500	76300	0.034	-0.178	0.181	0.793	
14	1179.0	3180.0	100900	76200	-0.442	-0.093	0.452	0.810	
15	2209.0	3182.0	101000	76200	0.131	-0.160	0.206	0.894	
16	4275.0	3188.0	101200	76200	0.697	-0.102	0.705	0.722	
17	5314.0	3191.0	101300	76200	0.401	-0.075	0.408	0.874	
18	6353.0	3193.0	101400	76200	0.105	-0.144	0.179	0.740	
19	7391.0	3196.0	101500	76200	-0.094	-0.117	0.150	0.823	
20	8437.0	3199.0	101600	76200	-1.066	-0.091	1.070	0.789	OUTLIER
21	1179.0	4215.0	100900	76100	-0.570	-0.321	0.654	0.806	
22	2209.0	4219.0	101000	76100	0.003	-0.195	0.195	0.863	
23	4273.0	4228.0	101200	76100	0.762	0.152	0.777	0.889	
24	5313.0	4233.0	101300	76100	0.369	0.372	0.524	0.862	
25	6352.0	4234.0	101400	76100	0.073	0.206	0.219	0.798	
26	7389.0	4236.0	101500	76100	-0.029	0.138	0.141	0.871	
27	8431.0	4241.0	101600	76100	-0.615	0.357	0.711	0.784	

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Affine:  X = 0.096531·px + 0.000123·ln + 100786.24
         Y = 0.000252·px + -0.096398·ln + 76506.34

RMS: 0.484 m   Max: 1.070 m   Threshold: 0.967 m
Grid: 8×4, spacing 100 m   Scale: 10.36×10.39 px/m
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Coordinate Transformation — EPSG:3152 → EPSG:3006

GCP	ST74 X	ST74 Y	SWEREF99TM E	SWEREF99TM N
1	101000	76400	674883.649	6578455.080
2	101100	76400	674983.542	6578459.669
3	101200	76400	675083.435	6578464.257
4	101300	76400	675183.328	6578468.846
5	101400	76400	675283.220	6578473.435
6	101500	76400	675383.113	6578478.024
7	100900	76300	674788.345	6578350.598
8	101000	76300	674888.238	6578355.187
9	101100	76300	674988.131	6578359.776
10	101200	76300	675088.024	6578364.365
11	101300	76300	675187.916	6578368.953
12	101400	76300	675287.809	6578373.542
13	101500	76300	675387.702	6578378.131
14	100900	76200	674792.934	6578250.706
15	101000	76200	674892.827	6578255.294
16	101200	76200	675092.612	6578264.472
17	101300	76200	675192.505	6578269.061
18	101400	76200	675292.398	6578273.649
19	101500	76200	675392.291	6578278.238
20	101600	76200	675492.184	6578282.827
21	100900	76100	674797.523	6578150.813
22	101000	76100	674897.415	6578155.402
23	101200	76100	675097.201	6578164.579
24	101300	76100	675197.094	6578169.168
25	101400	76100	675296.987	6578173.756
26	101500	76100	675396.880	6578178.345
27	101600	76100	675496.773	6578182.934

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Source CRS:  EPSG:3152  —  ST74, Stockholm 1938 (local TM)
Target CRS:  EPSG:3006  —  SWEREF99 TM

Pipeline:
  proj=pipeline step inv proj=tmerc lat_0=0 lon_0=18.05779 k=0.99999425 x_0=100178.1808 y_0=-6500614.7836 ellps=GRS80 step proj=utm zone=33 ellps=GR

Equivalent to Lantmäteriet GTRANS.
  1. Inverse ST74 TM → geographic on GRS80
  2. Forward UTM zone 33 → SWEREF99TM
No datum shift needed. Round-trip accuracy < 0.001 mm.
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